

Q.21 Explain the working of a piezoelectric transducer.
(CO1)

Q.22 How do bio-sensors function? Give an example.
(CO5)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Explain the construction, working, and applications of MEMS pressure sensors. (CO5)

Q.24 Describe the different types of actuators. Discuss the advantages and disadvantages of hydraulic and pneumatic actuators. (CO4)

Q.25 Discuss in detail the classification of temperature sensors and their applications in industry. (CO3)

(40)

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222843/212843

No. of Printed Pages : 4
Roll No.

222843/212843

4th Sem. / Automation & Robotics

Subject : Sensors and Actuators

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Which of the following is a temperature sensor?
(CO3)

- a) Thermocouple b) Capacitive Sensor
- c) Hall effect sensor d) Proximity switch

Q.2 A capacitive proximity sensor detects: (CO2)

- a) Temperature b) Metal objects
- c) Non-metallic objects
- d) Light

Q.3 Which transducer uses the piezoelectric effect?

- a) Thermoelectric transducer (CO1)
- b) Variable inductance transducer
- c) Piezoelectric transducer
- d) Photoelectric transducer

(1)

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Q.4 The working principle of a rotameter is based on: (CO3)

- a) Pressure drop b) Magnetic field
- c) Induced EMF d) Change in resistance

Q.5 Which actuator is driven by hydraulic fluid? (CO4)

- a) Solenoid b) Rotary actuator
- c) Pneumatic actuator d) Hydraulic actuator

Q.6 Smart sensors include: (CO5)

- a) Capacitive sensors
- b) Piezoelectric sensors
- c) Nono-sensors
- d) Inductive sensors

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 The principle of a Hall effect sensor is based on _____. (CO1)

Q.8 A diaphragm pressure sensor is mainly used for measuring _____. (CO2)

Q.9 Thermistors are used to measure _____. (CO3)

Q.10 MEMS sensors are used in _____ applications. (CO5)

Q.11 A _____ is used to measure liquid level using capacitance. (CO3)

Q.12 The main function of a solenoid is to _____. (CO4)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Classify different types of sensors and provide examples. (CO1)

Q.14 Explain the working of an inductive proximity sensor. (CO2)

Q.15 Describe the construction and working of a thermocouple. (CO3)

Q.16 What are resistance temperature detectors (RTD)? How do they work? (CO3)

Q.17 Differentiate between diaphragm and Bourdon tube pressure sensors. (CO2)

Q.18 How is flow measured using a Venturi tube? (CO3)

Q.19 Describe the working principle of a rotary actuator. (CO4)

Q.20 What is the role of smart sensors in modern industrial applications? (CO5)